Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9975369458128078	0.995085995085995 – 1.002469135802469

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0004237364001505	0.9950299033165496 – 1.0027191413237926

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9988765885739697	0.9974782263525346 – 1.00074901107132

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0009046751669703	0.9990133896044252 – 1.0019451996064153

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0025442864866585	0.997745055627488 – 1.0037790825210626

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1	0.9977220956719818 – 1.0045766590389016

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9996209159618072	0.9968862813218238 – 1.003373351119016

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0001749760866014	0.9966533344216799 – 1.0017720987711893

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	0.9979209979209976 – 1.0020876826722338

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear

CURRENT / BASELINE	1.0066889632107023	95% INTERVAL	1 – 1.0067114093959733

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.0070253462053007	1.0043716947147794 – 1.010413578148842

Cholesky factorization

256 × 256 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9936900729098042	0.9904663726385025 – 0.9961985447269444

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9949989260086665	0.9933922773224124 – 0.9990776670261271

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0075606057275983	0.9775864660039932 – 1.0349407616669952

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9969792279009496	0.9902385219515536 – 1.0019951031910102

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0001844035569398	0.9954224992336773 – 1.0014575763174642